

Concept 4 | Higher-Degree Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Factor theorem

$$P(a) = 0 \Rightarrow x - a \text{ is a factor}$$

Degree

$$\deg(PQ) = \deg P + \deg Q$$

Conjugate roots

$$a + bi \Rightarrow a - bi$$

Worked Solutions

Higher-Degree Equations

Q001 Written

$$x^3 - 8 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Q002 Written

$$x^3 = -1$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$.

Final answer

$$x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q003 Written

$$x^3 = 27$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$.

Final answer

$$x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$$

Q004 Written

$$x^3 = -125$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$.

Final answer

$$x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q005 Written

$$8x^3 = 1$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$.

Final answer

$$x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$$

Q006 MCQ

$$x^3 - 8 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Correct option: B

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Worked Solutions

Higher-Degree Equations

Q007 MCQ

 $x^3 = -1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$.

Correct option: D

Final answer

$$x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$$

Q008 MCQ

 $x^3 = 27$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$.

Correct option: B

Final answer

$$x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q009 MCQ

 $x^3 = -125$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$.

Correct option: A

Final answer

$$x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$$

Q010 MCQ

 $8x^3 = 1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$.

Correct option: C

Final answer

$$x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$$

Worked Solutions

Higher-Degree Equations

Q011 Written

$$x^3 - 2x^2 - 5x + 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 1, 3$.

Final answer

$$x = -2, 1, 3$$

Q012 Written

$$3x^3 + x^2 - 8x + 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 2/3, 1$.

Final answer

$$x = -2, 2/3, 1$$

Worked Solutions

Higher-Degree Equations

Q013 Written

$$2x^3 - 3x^2 - x - 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$.

Final answer

$$x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$$

Q014 Written

$$x^3 - x^2 - 16x - 20 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2(\text{repeated}), 5$.

Final answer

$$x = -2(\text{repeated}), 5$$

Worked Solutions

Higher-Degree Equations

Q015 Written

$$x^3 + 6x^2 + 11x + 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, -1$.

Final answer

$$x = -3, -2, -1$$

Q016 Written

$$x^3 + 3x^2 - 4x - 12 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, 2$.

Final answer

$$x = -3, -2, 2$$

Worked Solutions

Higher-Degree Equations

Q017 Written

$$3x^3 - 2x^2 - 3x + 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 2/3, 1$.

Final answer

$$x = -1, 2/3, 1$$

Q018 Written

$$x^3 - x^2 - 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$.

Final answer

$$x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q019 Written

$$x^3 + 4x^2 + 2x - 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Final answer

$$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$$

Q020 Written

$$x^3 - 4x^2 + 9x - 10 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, 1 - 2i, 1 + 2i$.

Final answer

$$x = 2, 1 - 2i, 1 + 2i$$

Worked Solutions

Higher-Degree Equations

Q021 Written

$$x^3 + x^2 - 8x - 12 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2$ (repeated), 3.

Final answer

$$x = -2(\text{repeated}), 3$$

Q022 Written

$$2x^3 + 7x^2 + 8x + 3 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1$ (repeated), $-3/2$.

Final answer

$$x = -1(\text{repeated}), -3/2$$

Worked Solutions

Higher-Degree Equations

Q023 Written

$$x^3 + 6x^2 + 12x + 8 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2$ (triple).

Final answer

$$x = -2(\text{triple})$$

Q024 MCQ

$$x^3 - 2x^2 - 5x + 6 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, 1, 3$.

Correct option: C

Final answer

$$x = -2, 1, 3$$

Worked Solutions

Higher-Degree Equations

Q025 MCQ

 $3x^3 + x^2 - 8x + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, 2/3, 1$.

Correct option: D

Final answer

$$x = -2, 2/3, 1$$

Q026 MCQ

 $2x^3 - 3x^2 - x - 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$.

Correct option: D

Final answer

$$x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$$

Worked Solutions

Higher-Degree Equations

Q027 MCQ

 $x^3 - x^2 - 16x - 20 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2$ (repeated), 5.

Correct option: B

Final answer $x = -2$ (repeated), 5

Q028 MCQ

 $x^3 + 6x^2 + 11x + 6 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, -2, -1$.

Correct option: C

Final answer $x = -3, -2, -1$

Worked Solutions

Higher-Degree Equations

Q029 MCQ

 $x^3 + 3x^2 - 4x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, -2, 2$.

Correct option: D

Final answer

$$x = -3, -2, 2$$

Q030 MCQ

 $3x^3 - 2x^2 - 3x + 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, 2/3, 1$.

Correct option: D

Final answer

$$x = -1, 2/3, 1$$

Worked Solutions

Higher-Degree Equations

Q031 MCQ

 $x^3 - x^2 - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$.

Correct option: A

Final answer

$$x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q032 MCQ

 $x^3 + 4x^2 + 2x - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Correct option: D

Final answer

$$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$$

Worked Solutions

Higher-Degree Equations

Q033 MCQ

 $x^3 - 4x^2 + 9x - 10 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, 1 - 2i, 1 + 2i$.

Correct option: D

Final answer

$$x = 2, 1 - 2i, 1 + 2i$$

Q034 MCQ

 $x^3 + x^2 - 8x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2(\text{repeated}), 3$.

Correct option: D

Final answer

$$x = -2(\text{repeated}), 3$$

Worked Solutions

Higher-Degree Equations

Q035 MCQ

 $2x^3 + 7x^2 + 8x + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1$ (repeated), $-3/2$.

Correct option: B**Final answer** $x = -1$ (repeated), $-3/2$

Q036 MCQ

 $x^3 + 6x^2 + 12x + 8 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2$ (triple).

Correct option: D**Final answer** $x = -2$ (triple)

Worked Solutions

Higher-Degree Equations

Q037 Written

$$x^4 - 7x^2 - 18 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$.

Final answer

$$x = -3, 3, -\sqrt{2}i, \sqrt{2}i$$

Q038 Written

$$x^4 - 5x^2 + 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1, 1, 2$.

Final answer

$$x = -2, -1, 1, 2$$

Worked Solutions

Higher-Degree Equations

Q039 Written

$$x^4 - 5x^2 - 36 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -2i, 2i$.

Final answer

$$x = -3, 3, -2i, 2i$$

Q040 Written

$$x^4 - 6x^2 - 27 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$.

Final answer

$$x = -3, 3, -\sqrt{3}i, \sqrt{3}i$$

Worked Solutions

Higher-Degree Equations

Q041 Written

$$x^4 - x^2 - 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -\sqrt{2}, \sqrt{2}, -i, i$.

Final answer

$$x = -\sqrt{2}, \sqrt{2}, -i, i$$

Q042 MCQ

$$x^4 - 7x^2 - 18 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$.

Correct option: D

Final answer

$$x = -3, 3, -\sqrt{2}i, \sqrt{2}i$$

Worked Solutions

Higher-Degree Equations

Q043 MCQ

 $x^4 - 5x^2 + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, -1, 1, 2$.

Correct option: D**Final answer**

$$x = -2, -1, 1, 2$$

Q044 MCQ

 $x^4 - 5x^2 - 36 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, 3, -2i, 2i$.

Correct option: C**Final answer**

$$x = -3, 3, -2i, 2i$$

Worked Solutions

Higher-Degree Equations

Q045 MCQ

 $x^4 - 6x^2 - 27 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$.

Correct option: C

Final answer

$$x = -3, 3, -\sqrt{3}i, \sqrt{3}i$$

Q046 MCQ

 $x^4 - x^2 - 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -\sqrt{2}, \sqrt{2}, -i, i$.

Correct option: B

Final answer

$$x = -\sqrt{2}, \sqrt{2}, -i, i$$

Worked Solutions

Higher-Degree Equations

Q047 Written

$$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$.

Final answer

$$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$$

Q048 Written

$$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3, 4$.

Final answer

$$x = 1, 2, 3, 4$$

Worked Solutions

Higher-Degree Equations

Q049 Written

$$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3(\text{repeated})$.

Final answer

$$x = 1, 2, 3(\text{repeated})$$

Q050 Written

$$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 3, 1 - i, 1 + i$.

Final answer

$$x = -1, 3, 1 - i, 1 + i$$

Worked Solutions

Higher-Degree Equations

Q051 MCQ

 $x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$.

Correct option: B

Final answer

$$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$$

Worked Solutions

Higher-Degree Equations

Q052 MCQ

 $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 1, 2, 3, 4$.

Correct option: A

Final answer $x = 1, 2, 3, 4$

Q053 MCQ

 $x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 1, 2, 3(\text{repeated})$.

Correct option: C

Final answer $x = 1, 2, 3(\text{repeated})$

Worked Solutions

Higher-Degree Equations

Q054 MCQ

 $x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, 3, 1 - i, 1 + i$.

Correct option: C

Final answer

$$x = -1, 3, 1 - i, 1 + i$$

Q055 Written

 $x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 1, b = -6$; other roots: -2 .

Final answer

$$a = 1, b = -6; \text{ other roots: } -2$$

Worked Solutions

Higher-Degree Equations

Q056 Written

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a, b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 2, b = 2$; other roots: 1.

Final answer

$a = 2, b = 2$; other roots: 1

Q057 Written

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a, b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -7, b = 14$; other roots: 4.

Final answer

$a = -7, b = 14$; other roots: 4

Worked Solutions

Higher-Degree Equations

Q058 MCQ

$x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 1, b = -6$; other roots: -2 .

Correct option: C

Final answer

$a = 1, b = -6$; other roots: -2

Worked Solutions

Higher-Degree Equations

Q059 MCQ

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a, b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 2, b = 2$; other roots: 1.

Correct option: B

Final answer

$a = 2, b = 2$; other roots: 1

Worked Solutions

Higher-Degree Equations

Q060 MCQ

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a, b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -7, b = 14$; other roots: 4.

Correct option: C

Final answer

$a = -7, b = 14$; other roots: 4

Worked Solutions

Higher-Degree Equations

Q061 Written

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -3, b = 1$; other roots: $-1, 2 - i$.

Final answer

$a = -3, b = 1$; other roots: $-1, 2 - i$

Q062 Written

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 6$; other roots: $1 - i, 2$.

Final answer

$a = -4, b = 6$; other roots: $1 - i, 2$

Worked Solutions

Higher-Degree Equations

Q063 Written

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 14$; other roots: $1 + 3i, 2$.

Final answer

$a = -4, b = 14$; other roots: $1 + 3i, 2$

Q064 Written

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 0, b = 1$; other roots: $1 - 2i, -2$.

Final answer

$a = 0, b = 1$; other roots: $1 - 2i, -2$

Worked Solutions

Higher-Degree Equations

Q065 MCQ

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -3, b = 1$; other roots: $-1, 2 - i$.

Correct option: A

Final answer

$a = -3, b = 1$; other roots: $-1, 2 - i$

Q066 MCQ

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -4, b = 6$; other roots: $1 - i, 2$.

Correct option: B

Final answer

$a = -4, b = 6$; other roots: $1 - i, 2$

Worked Solutions

Higher-Degree Equations

Q067 MCQ

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -4, b = 14$; other roots: $1 + 3i, 2$.

Correct option: D

Final answer

$a = -4, b = 14$; other roots: $1 + 3i, 2$

Q068 MCQ

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 0, b = 1$; other roots: $1 - 2i, -2$.

Correct option: B

Final answer

$a = 0, b = 1$; other roots: $1 - 2i, -2$

Worked Solutions

Higher-Degree Equations

Q069 Written

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $k = 9$; other roots: 3, 4.

Final answer

$k = 9$; other roots: 3, 4

Q070 MCQ

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $k = 9$; other roots: 3, 4.

Correct option: B

Final answer

$k = 9$; other roots: 3, 4

Worked Solutions

Higher-Degree Equations

Q071 Written

For an imaginary cube root of unity w , evaluate w^9 .**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q072 Written

For an imaginary cube root of unity w , evaluate w^{2010} .**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Worked Solutions

Higher-Degree Equations

Q073 Written

For an imaginary cube root of unity w , evaluate w^3 .

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q074 MCQ

For an imaginary cube root of unity w , evaluate w^9 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: A

Final answer

1

Worked Solutions

Higher-Degree Equations

Q075 MCQ

For an imaginary cube root of unity w , evaluate w^{2010} Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: B**Final answer**

1

Q076 MCQ

For an imaginary cube root of unity w , evaluate w^3 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: D**Final answer**

1

Worked Solutions

Higher-Degree Equations

Q077 Written

For an imaginary cube root of unity w , evaluate $w^4 + w^5$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer

-1

Q078 Written

For an imaginary cube root of unity w , evaluate $w^2 + w$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer

-1

Worked Solutions

Higher-Degree Equations

Q079 Written

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q080 MCQ

For an imaginary cube root of unity w , evaluate $w^4 + w^5$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1 .

Correct option: A

Final answer -1

Worked Solutions

Higher-Degree Equations

Q081 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1 .

Correct option: B

Final answer -1

Q082 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 0 .

Correct option: D

Final answer 0

Worked Solutions

Higher-Degree Equations

Q083 Written

For an imaginary cube root of unity w , evaluate $w + 1/w$.**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer -1

Q084 Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer 1

Worked Solutions

Higher-Degree Equations

Q085 Written

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q086 MCQ

For an imaginary cube root of unity w , evaluate $w + 1/w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1 .

Correct option: D

Final answer -1

Worked Solutions

Higher-Degree Equations

Q087 MCQ

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: C**Final answer**

1

Q088 MCQ

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 0.

Correct option: D**Final answer**

0

Worked Solutions

Higher-Degree Equations

Q089 Written

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -4$ or $9/4$.

Final answer

$a = -4$ or $9/4$

Q090 Written

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -3$ or 1 .

Final answer

$a = -3$ or 1

Worked Solutions

Higher-Degree Equations

Q091 **Written** $x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root.**Solution**

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = 5$ or 9 .

Final answer $a = 5$ or 9 Q092 **Written** $x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root.**Solution**

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -6, -2$, or 10 .

Final answer $a = -6, -2$, or 10

Worked Solutions

Higher-Degree Equations

Q093 Written

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -7, -3$, or 2 .

Final answer $a = -7, -3$, or 2

Q094 MCQ

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -4$ or $9/4$.

Correct option: D

Final answer $a = -4$ or $9/4$

Worked Solutions

Higher-Degree Equations

Q095 MCQ

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -3$ or 1 .

Correct option: D

Final answer

$a = -3$ or 1

Q096 MCQ

$x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 5$ or 9 .

Correct option: B

Final answer

$a = 5$ or 9

Worked Solutions

Higher-Degree Equations

Q097 MCQ

$x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -6, -2$, or 10 .

Correct option: A

Final answer

$a = -6, -2$, or 10

Q098 MCQ

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -7, -3$, or 2 .

Correct option: A

Final answer

$a = -7, -3$, or 2

Worked Solutions

Higher-Degree Equations

Q099 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$.

Final answer

$a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$

Worked Solutions

Higher-Degree Equations

Q100 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion.

Solution

1. Use $(w^3 = 1)$ and reduce powers modulo 3.
2. Use $(1 + w + w^2 = 0)$ to simplify the remaining expression.
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions

Higher-Degree Equations

Q101 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = 2/3, 2$, or 6 .

Final answer

$a = 2/3, 2$, or 6

Worked Solutions

Higher-Degree Equations

Q102 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$.

Solution

1. Set inflow equal to outflow: $(x^4 + x^2 + x = x^3 + 2)$.
2. Move all terms to one side.
3. Therefore the final answer is $x^4 - x^3 + x^2 + x - 2 = 0$.

Final answer

$$x^4 - x^3 + x^2 + x - 2 = 0$$

Worked Solutions

Higher-Degree Equations

Q103 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ **for the stable pressure levels.**

Solution

1. Factor $(x^4 - x^3 + x^2 + x - 2)$ and solve the remaining quadratic.
2. Apply the stated physical restriction ($x \geq 0$) only after listing the algebraic roots.
3. Therefore the final answer is $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$.

Final answer

$$x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q104 Challenge

Classify the stable pressure solutions as real or complex.

Solution

1. Separate the real roots from the non-real conjugate pair.

Both types occur: real $x = -1, 1$

2. Therefore the final answer is non-real conjugates $x = (1 \pm \sqrt{7}i)/2$.

if $x \geq 0 : x = 1$

Final answer

Both types occur: real $x = -1, 1$

non-real conjugates $x = (1 \pm \sqrt{7}i)/2$

if $x \geq 0 : x = 1$

Worked Solutions

Higher-Degree Equations

Q105 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$.

Correct option: C

Final answer

$a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$

Worked Solutions

Higher-Degree Equations

Q106 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 4.

Correct option: D

Final answer

4

Worked Solutions

Higher-Degree Equations

Q107 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 2/3, 2$, or 6 .

Correct option: D

Final answer

$a = 2/3, 2$, or 6

Worked Solutions

Higher-Degree Equations

Q108 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x^4 - x^3 + x^2 + x - 2 = 0$.

Correct option: A

Final answer

$$x^4 - x^3 + x^2 + x - 2 = 0$$

Worked Solutions

Higher-Degree Equations

Q109 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ **for the stable pressure levels Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$.

Correct option: D

Final answer

$$x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q110 Challenge

Classify the stable pressure solutions as real or complex Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.

Both types occur: real $x = -1, 1$

2. The correct value is non-real conjugates $x = (1 \pm \sqrt{7}i)/2$.

if $x \geq 0 : x = 1$

Correct option: B

Final answer

Both types occur: real $x = -1, 1$

non-real conjugates $x = (1 \pm \sqrt{7}i)/2$

if $x \geq 0 : x = 1$

Worked Solutions

Higher-Degree Equations

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $x^3 - 8 = 0$. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Correct option: D

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Quiz MCQ 2 **Concept Quiz · MCQ**

Are $1 + 2i$ and $1 - 2i$ roots of $x^2 - 2x + 5 = 0$? Which root must also occur? Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $1 - 2i$.

Correct option: D

Final answer

$$1 - 2i$$

Worked Solutions

Higher-Degree Equations

Quiz MCQ 3 **Concept Quiz · MCQ**

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1 .

Correct option: B

Final answer

-1

Quiz MCQ 4 **Concept Quiz · MCQ**

For the remaining quadratic factor, a repeated root requires Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $D = 0$.

Correct option: B

Final answer

$D = 0$

Worked Solutions

Higher-Degree Equations

Quiz WRITTEN 5 Concept Quiz · Written

$$\text{Solve } x^4 - 5x^2 + 4 = 0.$$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $x = -2, -1, 1, 2$

Final answer

$$x = -2, -1, 1, 2$$

Quiz WRITTEN 6 Concept Quiz · Written

$$\text{For } x^3 + ax^2 + bx + 10 = 0 \text{ with root } 1 + 2i, \text{ find } a, b \text{ and the other roots.}$$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = 0, b = 1$; other roots: $1 - 2i, -2$

Final answer

$$a = 0, b = 1; \text{ other roots: } 1 - 2i, -2$$

Worked Solutions

Higher-Degree Equations

Quiz WRITTEN 7 Concept Quiz · Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: 1

Final answer

1

Quiz WRITTEN 8 Concept Quiz · Written

Find all real a for $x^3 - 2x^2 + (a - 3)x + a = 0$ to have a repeated root.

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = -4$ or $9/4$

Final answer

$a = -4$ or $9/4$

Answer key

Use the question number shown on each page.

Q001 $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$	Q027 B	Q053 C	Q079 0
Q002 $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$	Q028 C	Q054 C	Q080 A
Q003 $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$	Q029 D	Q055 $a = 1, b = -6$; other roots: -2	Q081 B
Q004 $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$	Q030 D	Q056 $a = 2, b = 2$; other roots: 1	Q082 D
Q005 $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$	Q031 A	Q057 $a = -7, b = 14$; other roots: 4	Q083 -1
Q006 B	Q032 D	Q058 C	Q084 1
Q007 D	Q033 D	Q059 B	Q085 0
Q008 B	Q034 D	Q060 C	Q086 D
Q009 A	Q035 B	Q061 $a = -3, b = 1$; other roots: $-1, 2 - i$	Q087 C
Q010 C	Q036 D	Q062 $a = -4, b = 6$; other roots: $1 - i, 2$	Q088 D
Q011 $x = -2, 1, 3$	Q037 $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$	Q063 $a = -4, b = 14$; other roots: $1 + 3i, 2$	Q089 $a = -4$ or $9/4$
Q012 $x = -2, 2/3, 1$	Q038 $x = -2, -1, 1, 2$	Q064 $a = 0, b = 1$; other roots: $1 - 2i, -2$	Q090 $a = -3$ or 1
Q013 $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$	Q039 $x = -3, 3, -2i, 2i$	Q065 A	Q091 $a = 5$ or 9
Q014 $x = -2$ (repeated), 5	Q040 $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$	Q066 B	Q092 $a = -6, -2$, or 10
Q015 $x = -3, -2, -1$	Q041 $x = -\sqrt{2}, \sqrt{2}, -i, i$	Q067 D	Q093 $a = -7, -3$, or 2
Q016 $x = -3, -2, 2$	Q042 D	Q068 B	Q094 D
Q017 $x = -1, 2/3, 1$	Q043 D	Q069 $k = 9$; other roots: $3, 4$	Q095 D
Q018 $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$	Q044 C	Q070 B	Q096 B
Q019 $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$	Q045 C	Q071 1	Q097 A
Q020 $x = 2, 1 - 2i, 1 + 2i$	Q046 B	Q072 1	Q098 A
Q021 $x = -2$ (repeated), 3	Q047 $x = 2$ (repeated), $1 - \sqrt{5}, 1 + \sqrt{5}$	Q073 1	Q099 $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$
Q022 $x = -1$ (repeated), $-3/2$	Q048 $x = 1, 2, 3, 4$	Q074 A	Q100 4
Q023 $x = -2$ (triple)	Q049 $x = 1, 2, 3$ (repeated)	Q075 B	Q101 $a = 2/3, 2$, or 6
Q024 C	Q050 $x = -1, 3, 1 - i, 1 + i$	Q076 D	Q102 $x^4 - x^3 + x^2 + x - 2 = 0$
Q025 D	Q051 B	Q077 -1	Q103 $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$
Q026 D	Q052 A	Q078 -1	

Both types occur: real $x = -1, 1$

Q104 non-real conjugates $x = (1 \pm \sqrt{7}i)/2$
if $x \geq 0 : x = 1$

Q105 C

Q106 D

Q107 D

Q108 A

Q109 D

Q110 B

Quiz MCQ 1 D

Quiz MCQ 2 D

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $x = -2, -1, 1, 2$

Quiz WRITTEN 6 $a = 0, b = 1$; other roots: $1 - 2i, -2$

Quiz WRITTEN 7 1

Quiz WRITTEN 8 $a = -4$ or $9/4$